

Name: _____ Date: _____

Period: _____

Exit Ticket — Answer independently

1. Solve $\sin \theta = -\frac{\sqrt{3}}{2}$ for all θ in the interval $[0, 2\pi]$.

Step 1 — Identify which quadrants have **negative** sine: _____

Step 2 — Reference angle: $\arcsin\left(\frac{\sqrt{3}}{2}\right) =$ _____

Step 3 — Write both solutions in $[0, 2\pi]$:

$\theta =$ _____ and $\theta =$ _____

2. Explain in one or two sentences why a trigonometric equation has *infinitely many* solutions when the domain is all real numbers.
